# Inho Choi

Ph.D. Candidate
National University of Singapore

Email : inhochoi@comp.nus.edu.sg
Website: https://ihchoi12.github.io/
Mobile : +65-89298595 (Singapore)
+82-1071547231 (Korea)


## Education

- **National University of Singapore** — Singapore
  *Ph.D. Candidate in School of Computing (Advisor: Jialin Li);* — *August 2019 – December 2025 (Expected)*

- **Yonsei University** — Seoul, Republic of Korea
  *Bachelor's Degree in Computer Science;* — *March 2012 – August 2019*

- **Uppsala University** — Uppsala, Sweden
  *Exchange Student in Information Technology;* — *August 2017 – January 2018*

## Experiences

- **Microsoft Research - PhD Research Intern** — Redmond, WA, USA
  *Systems Research Group (Mentor: Irene Zhang)* — *June 2025 — September 2025*
  - **ML-native Datapath Operating Systems**: Designing ML-based adaptive and automatic OS performance tuning.

- **Microsoft Research - PhD Research Intern** — Redmond, WA, USA
  *Systems Research Group (Mentor: Dan Ports)* — *June 2024 — August 2024*
  - **Applying Capybara (live TCP migration) for production systems**: Implemented live TCP migration system on Azure Web Cache cluster for seamless Redis server maintenance.

- **AMD - PhD Research Intern** — Singapore
  *Xilinx Lab - FPGA / System Design (Mentor: Guanwen Zhong)* — *May 2023 - August 2023*
  - **Hardware Accelerator in Data Centers**: Work on a research project of RDMA and computation offloading using FPGA-based SmartNIC

- **National University of Singapore - Research Intern** — Singapore
  *Systems & Network Security Lab (Advisor: Min Suk Kang)* — *September 2018 - February 2019*
  - **Blockchain Systems & Network Security**: Investigate partitioning attacks against Blockchain network.

- **Yonsei University - Research Intern** — Seoul, Korea
  *Dependable Computing Lab (Advisor: Kyoungwoo Lee)* — *February 2017 - May 2018*
  - **Sensing Systems using Deep Learning**: Design a realtime stress recognition system using deep learning.

- **Metlife - Summer Intern** — Seoul, Korea
  *IT Planning Team* — *July 2018 - August 2018*

## Work in Progress *(projects that I'm currently leading)*

- **ML-native Dataplane Operating Systems** [Current Main Project]:
  We design an ML-native dataplane OS architecture for automatic parameter tuning. Performance tuning remains a persistent challenge in modern datacenters, especially at microsecond scales. We are exploring how machine learning can be natively integrated into the OS dataplane, with case studies showing substantial performance gains through dynamic parameter optimization.

- **Capybara: Dynamic Layer-4 Load Balancing with Microsecond-Scale TCP Migration** [Under Review]:
  Layer-4 load balancers are a popular solution to high tail latencies but perform poorly under unpredictable workloads and traffic bursts because they statically assign connections to servers. We present Capybara, the first dynamic L4 load balancer with $\mu$s-scale stateful connection migration. Capybara leverages two trends – programmable switches and kernel-bypass – to efficiently implement TCP migration without packet loss, while maintaining transparency to clients.

## Publications

- **[APSYS '25] ML-native Dataplane Operating Systems**:
  **Inho Choi**, Anand Bonde, Jing Liu, Joshua Fried, Irene Zhang, Jialin Li.

  16th ACM SIGOPS Asia-Pacific Workshop on Systems (APSys 2025)

- **[ArXiv] A Primer on RecoNIC: RDMA-enabled Compute Offloading on SmartNIC**:
  Guanwen Zhong, Aditya Kolekar, Burin Amornpaisannon, **Inho Choi**, Haris Javaid, Mario Baldi.

  ArXiv, Dec, 2023

- **[APSYS '23] Capybara: $\mu$Second-scale live TCP migration**:
  **Inho Choi**, Nimish Wadekar, Raj Joshi, Joshua Fried, Dan R. K. Ports, Irene Zhang, and Jialin Li.

  14th ACM SIGOPS Asia-Pacific Workshop on Systems (APSys 2023)

- **[SIGCOMM '23] Network Load Balancing with In-network Reordering Support for RDMA**:
  Cha Hwan Song, Xin Zhe Khooi, Raj Joshi, **Inho Choi**, Jialin Li, and Mun Choon Chan.

  Proceedings of the 2023 ACM SIGCOMM Conference

- **[NSDI '23] Hydra: Serialization-Free Network Ordering for Strongly Consistent Distributed Applications**:
  **Inho Choi**, Ellis Michael, Yunfan Li, Dan Ports, and Jialin Li.

  Proceedings of the 20th USENIX Conference on Network Systems Design and Implementation

- **[S&P (Oakland) '20] A Stealthier Partitioning Attack against Bitcoin Peer-to-Peer Network**:
  Muoi Tran, **Inho Choi**, Gi Jun Moon, Viet-Anh Vu, and Min Suk Kang.

  In Proceedings of IEEE Symposium on Security and Privacy, May 2020.

- **[UbiComp Workshop '17] Multimodal Data Collection Framework for Mental Stress Monitoring**:
  Saewon Kye, Junhyung Moon, Juneil Lee, **Inho Choi**, Dongmi Cheon, Kyoungwoo Lee.

  In Proceedings of the 2017 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2017 ACM International Symposium on Wearable Computers. (Workshop Paper)

## Awards

- **Research Incentive Award** — October 2023
  *National University of Singapore*
- **NUS Research Scholarship** — August 2019 – July 2023
  *National University of Singapore*
- **Research Achievement Award** — January 2023
  *National University of Singapore*
- **Honors - 1st Semester, 2018** — August 2018
  *Yonsei University*

## Teaching Experiences

- **Teaching Assistant for Distributed Systems (CS5223)** — Semester 1, 2022
  *National University of Singapore*
- **Teaching Assistant for Introduction to Operating Systems (CS2106)** — Semester 1, 2021
  *National University of Singapore*
- **Teaching Assistant for Distributed Systems (CS5223)** — Semester 2, 2020
  *National University of Singapore*
- **Teaching Assistant for Programming Methodology (CS1010E)** — Semester 1, 2020
  *National University of Singapore*
- **Teaching Assistant for Cryptography Theory and Practice (CS4236)** — Semester 1, 2019
  *National University of Singapore*

## Mentoring Experiences

- **Yiyang Liu — NUS, Singapore** 2024
  *Master's Thesis: Enhancing Distributed Systems with Hydra: A Software Solution for Scalable Network Ordering*

## Services

- **Reviewer** ToN '25
- **Shadow PC Reviewer** EuroSys '26, EuroSys '25
- **Student Volunteer** SOSP '21

## Invited Talks

- **Co-Designing Systems for Microsecond-Scale Consistent Tail Latency in Datacenters**:

  Yonsei University (2025), Seoul National University (2025)